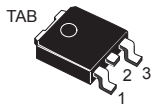
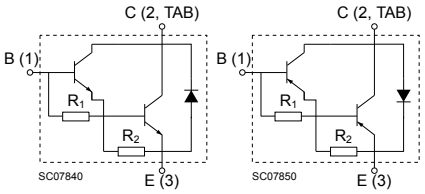
MJD122T4, MJD127T4

Datasheet

Complementary power Darlington transistors



DPAK



NPN: R_1 = 7 kΩ, R_2 = 70 Ω PNP: R_1 = 16 kΩ, R_2 = 60 Ω

STPOWER

Features

- Low collector-emitter saturation voltage
- Monolithic Darlington configuration with integrated antiparallel collector-emitter diode

Application

- Linear and switching industrial equipment

Description

These devices are manufactured in planar technology with “base island” layout and monolithic Darlington configuration.

Product status links
MJD122T4
MJD127T4

Product summary	
Order code	**MJD122T4**
Marking	MJD122
Polarity	NPN
Package	DPAK
Packing	Tape and reel
Order code	**MJD127T4**
Marking	MJD127
Polarity	PNP
Package	DPAK
Packing	Tape and reel

1 Electrical ratings

Table 1. Absolute maximum ratings

Symbol	Parameter	Value	Unit
V_{CEO}	Collector-emitter voltage (I_B = 0 A)	100	V
V_{CBO}	Collector-base voltage (I_E = 0 A)		V
V_{EBO}	Emitter-base voltage (I_C = 0 A)	5	V
I_C	Collector current	8	A
I_{CM}	Collector peak current	16	A
I_B	Base current	120	mA
P_{TOT}	Total power dissipation at T_C = 25 °C	20	W
T_{stg}	Storage temperature range	-65 to 150	°C
T_J	Operating junction temperature range		

Note: For PNP types voltage and current values are negative.

Table 2. Thermal data

Symbol	Parameter	Value	Unit
R_{thJC}	Thermal resistance, junction-to-case	6.25	°C/W

2 Electrical characteristics

T_C = 25°C unless otherwise specified.

Table 3. Electrical characteristics

Symbol	Parameter	Test conditions	Min.	Typ.	Max.	Unit
I_{CBO}	Collector cut-off current	V_{CB} = 100 V, I_E = 0 A		-	10	µA
I_{CEO}	Collector cut-off current	V_{CE} = 50 V, I_B = 0 A		-	10	µA
I_{EBO}	Emitter cut-off current	V_{EB} = 5 V, I_C = 0 A		-	2	mA
$V_{CEO(sus)}$ (1)	Collector-emitter sustaining voltage	I_C = 30 mA, I_B = 0 A	100	-		V
$V_{CE(sat)}$ (1)	Collector-emitter saturation voltage	I_C = 4 A, I_B = 16 mA		-	2	V
		I_C = 8 A, I_B = 80 mA		-	4	
$V_{BE(sat)}$ (1)	Base-emitter saturation voltage	I_C = 8 A, I_B = 80 mA		-	4.5	V
$V_{BE(on)}$	Base-emitter on voltage	I_C = 4 A, V_{CE} = 4 V		-	2.8	V
h_{FE} (1)	DC current gain	I_C = 4 A, V_{CE} = 4 V	1000	-	12000	
		I_C = 8 A, V_{CE} = 4 V	100	-		

1. *Pulse test: pulse duration ≤ 300 µs, duty cycle ≤ 2 %.*

Note: For PNP types voltage and current values are negative.

2.1 Electrical characteristics (curves)

Figure 1. Safe operating area

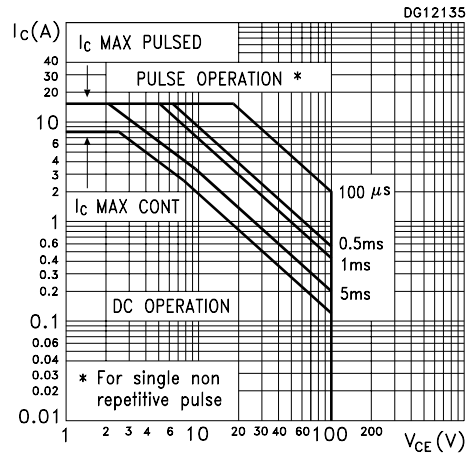


Figure 2. Derating curve

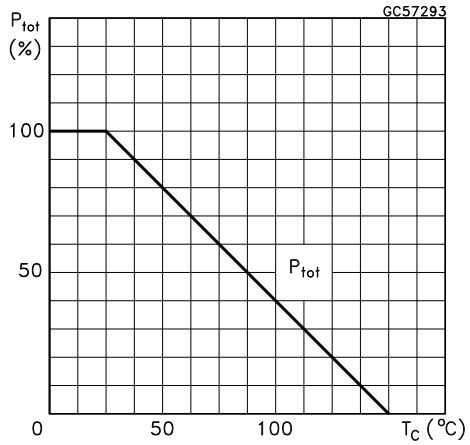


Figure 3. DC current gain (NPN)

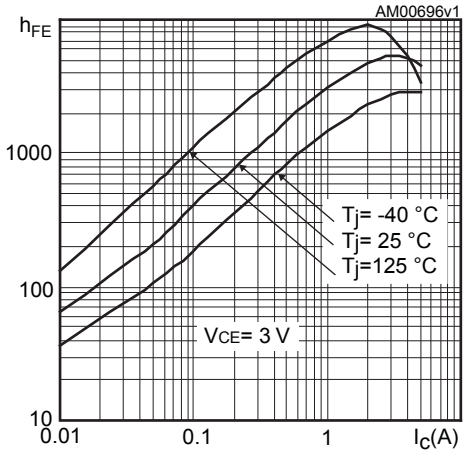


Figure 4. DC current gain (PNP)

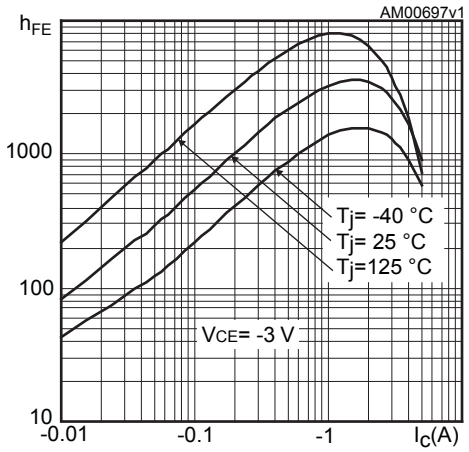


Figure 5. Collector-emitter saturation voltage (NPN)

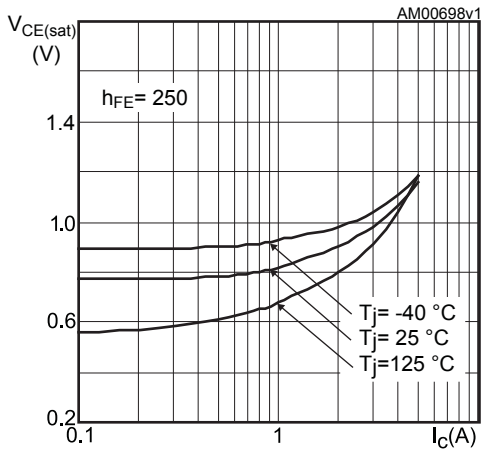


Figure 6. Collector-emitter saturation voltage (PNP)

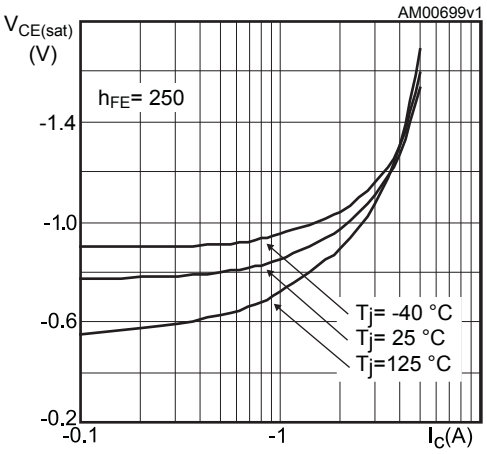


Figure 7. Base-emitter saturation voltage (NPN)

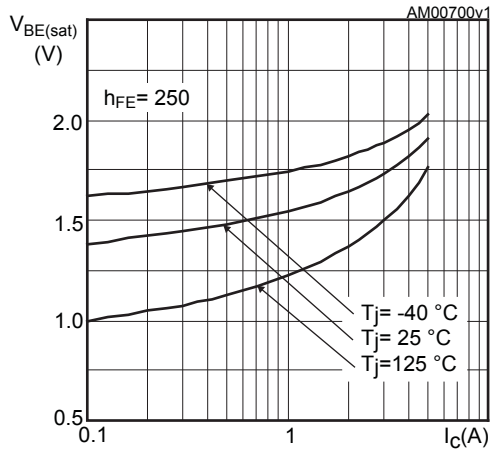


Figure 8. Base-emitter saturation voltage (PNP)

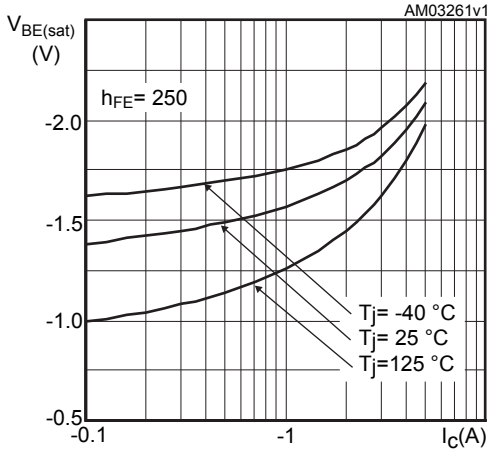


Figure 9. Base-emitter on voltage (NPN)

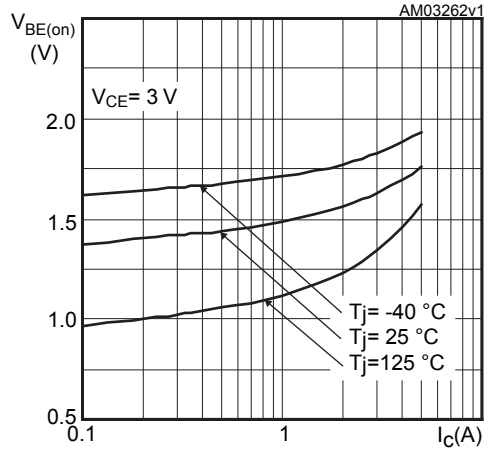


Figure 10. Base-emitter on voltage (PNP)

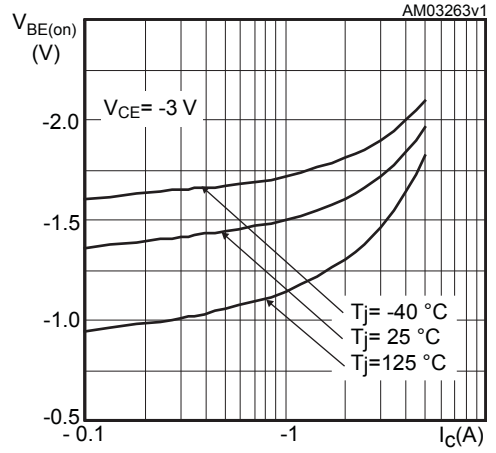


Figure 11. Resistive load switching time (NPN, on)

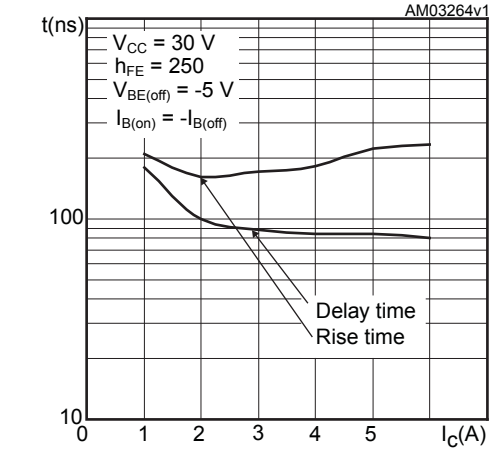


Figure 12. Resistive load switching time (PNP, on)

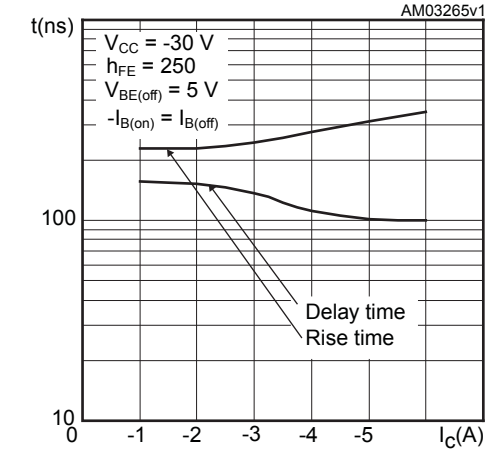


Figure 13. Resistive load switching time (NPN, off)

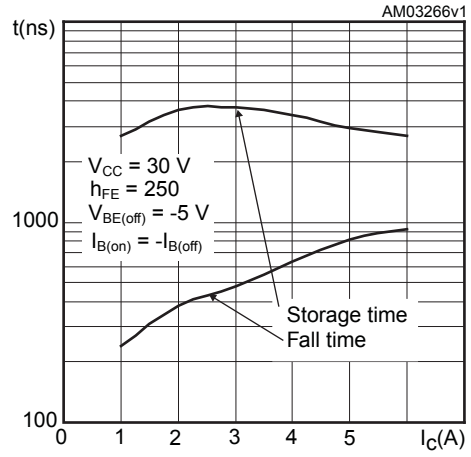


Figure 14. Resistive load switching time (PNP, off)

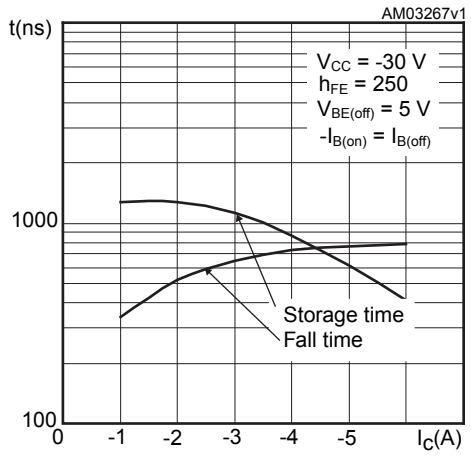


Figure 15. Capacitance characteristics (NPN)

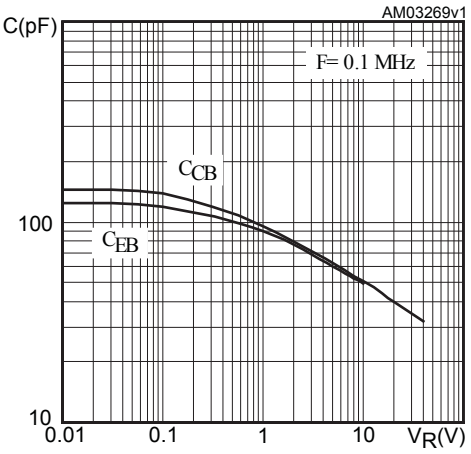
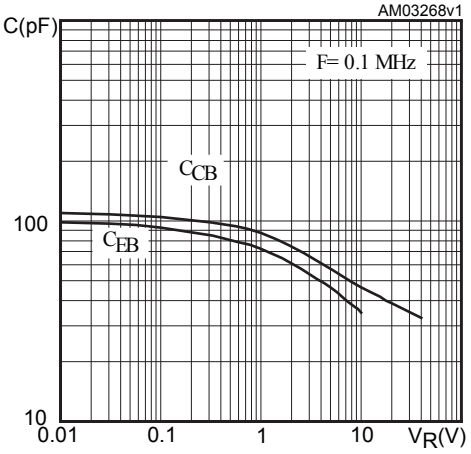
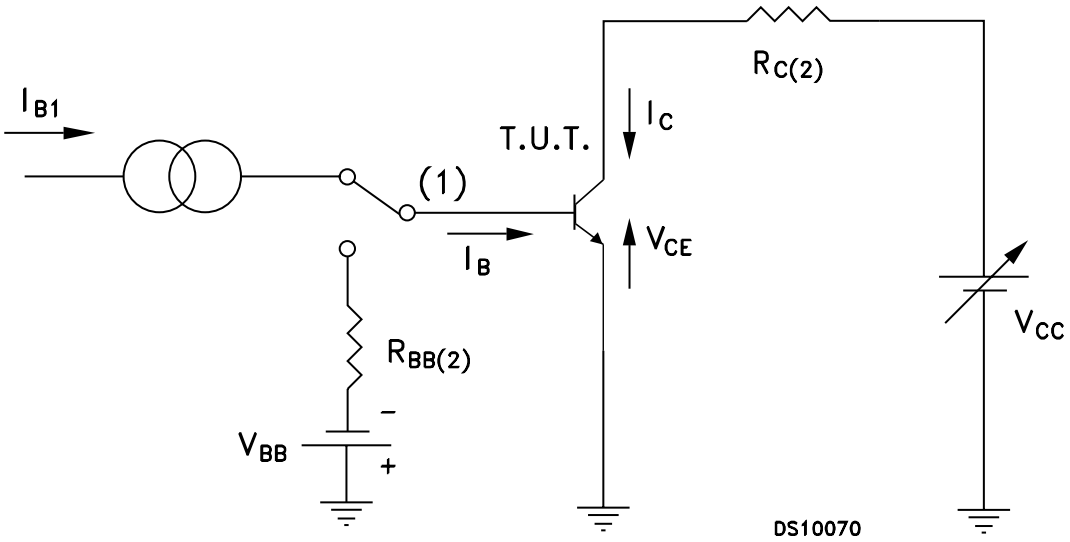


Figure 16. Capacitance characteristics (PNP)



3 Test circuits

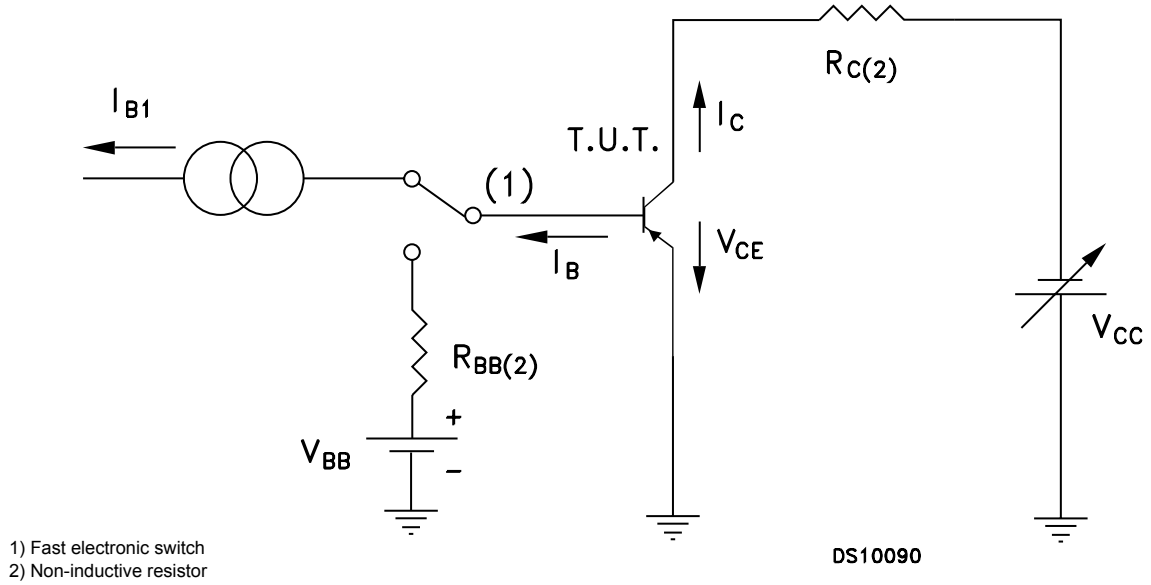
Figure 17. **Resistive load switching for NPN type**



Note: (1) Fast electronic switch.

Note: (2) Non-inductive resistor.

Figure 18. **Resistive load switching for PNP type**



Note: (1) Fast electronic switch.

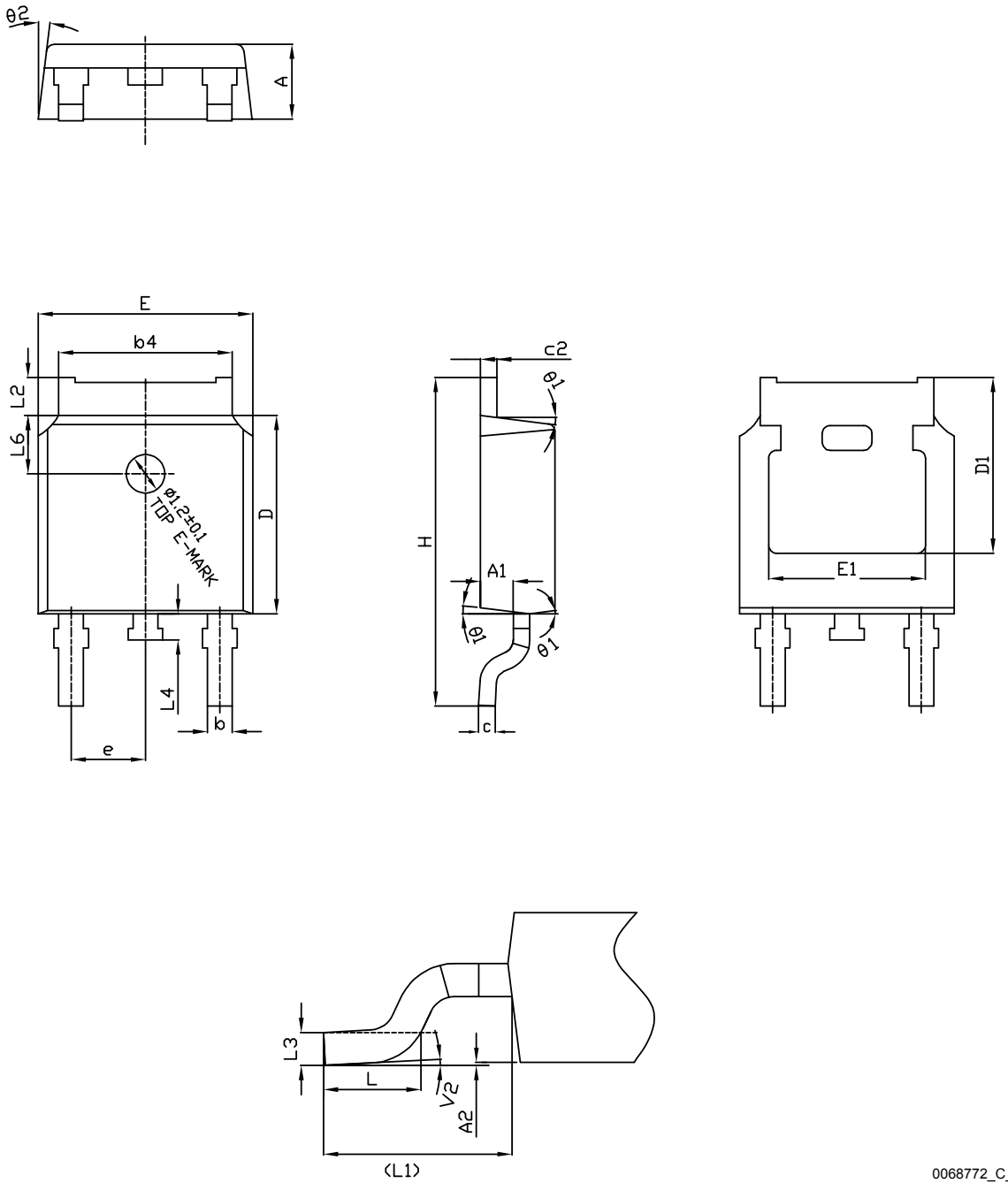
Note: (2) Non-inductive resistor.

4 Package information

To meet environmental requirements, ST offers these devices in different grades of ECOPACK packages, depending on their level of environmental compliance. ECOPACK specifications, grade definitions, and product status are available at: www.st.com. ECOPACK is an ST trademark.

4.1 DPAK (TO-252) type C package information

Figure 19. DPAK (TO-252) type C package outline

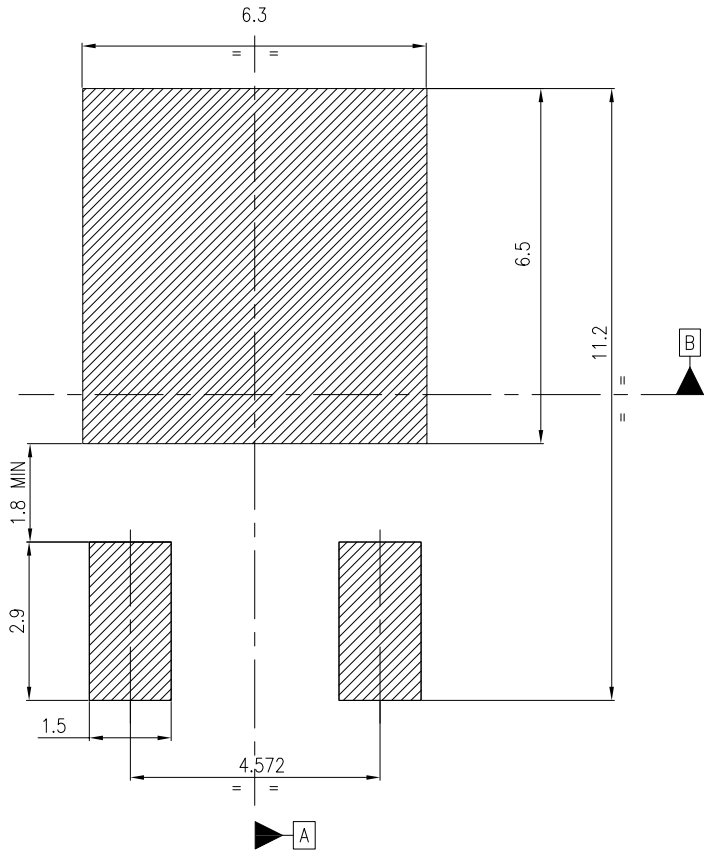


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Table 4. **DPAK (TO-252) type C mechanical data**

Dim.	mm		
	Min.	Typ.	Max.
A	2.20	2.30	2.38
A1	0.90	1.01	1.10
A2	0.00		0.10
b	0.72		0.85
b4	5.13	5.33	5.46
c	0.47		0.60
c2	0.47		0.60
D	6.00	6.10	6.20
D1	5.15	5.40	5.65
E	6.50	6.60	6.70
E1	4.70	4.85	5.00
e	2.186	2.286	2.386
H	9.80	10.10	10.40
L	1.40	1.50	1.70
L1		2.90 REF	
L2	0.90		1.25
L3		0.51 BSC	
L4	0.60	0.80	1.00
L6		1.80 BSC	
θ1	5°	7°	9°
θ2	5°	7°	9°
V2	0°		8°

Figure 20. DPAK (TO-252) recommended footprint (dimensions are in mm)



Notes:

1) This footprint is able to ensure insulation up to 630 Vrms (according to CEI IEC 664-1)

2) The device must be positioned within ⌖ 0.05 A B

FP_0068772_36

4.2 DPAK (TO-252) packing information

Figure 21. **DPAK (TO-252) tape outline**

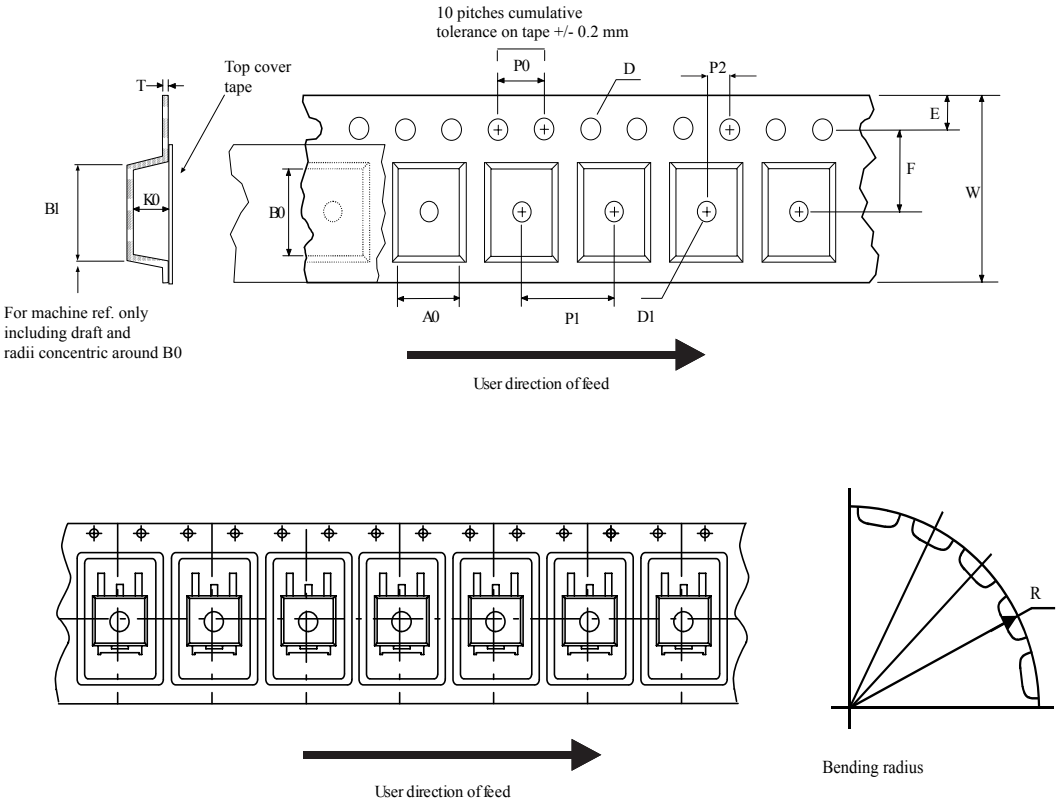
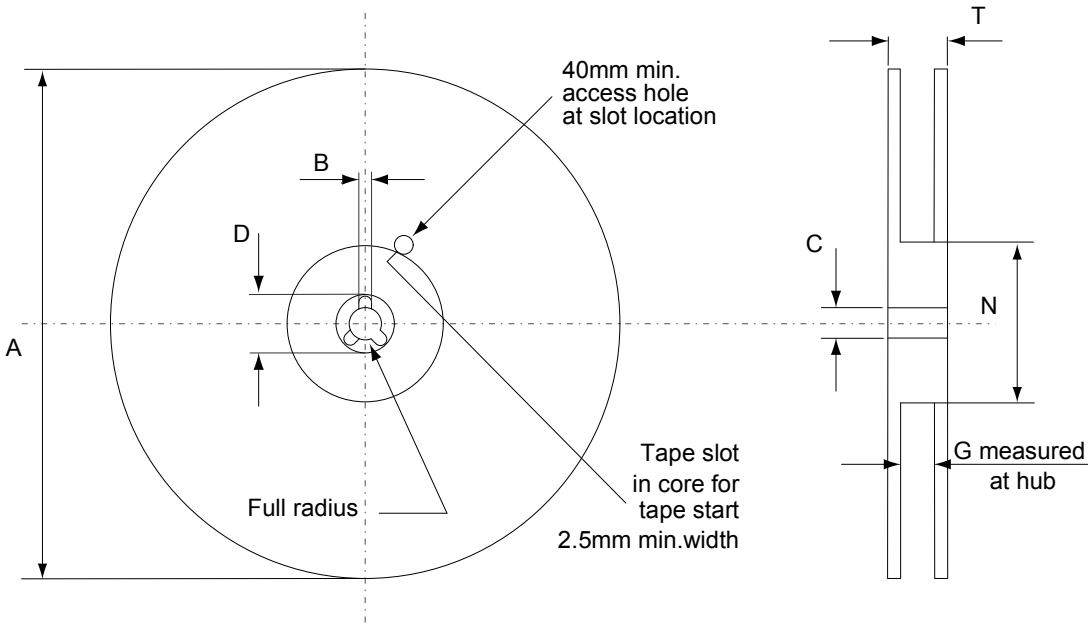


Figure 22. **DPAK (TO-252) reel outline**



AM06038v1

Table 5. **DPAK (TO-252) tape and reel mechanical data**

Tape			Reel		
Dim.	mm		Dim.	mm	
	Min.	Max.		Min.	Max.
A0	6.8	7	A		330
B0	10.4	10.6	B	1.5	
B1		12.1	C	12.8	13.2
D	1.5	1.6	D	20.2	
D1	1.5		G	16.4	18.4
E	1.65	1.85	N	50	
F	7.4	7.6	T		22.4
K0	2.55	2.75			
P0	3.9	4.1	Base qty.		2500
P1	7.9	8.1	Bulk qty.		2500
P2	1.9	2.1			
R	40				
T	0.25	0.35			
W	15.7	16.3			

Revision history

Table 6. **Document revision history**

Date	Revision	Changes
28-Apr-2025	1	First release.
16-Jun-2025	2	Updated Table 3. Electrical characteristics. Updated Section 4.1: DPAK (TO-252) type C package information.

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